

FIN ST1992–ST2081

Energy-Compatible Hodge Measure Classification,
Schur No-Go and the Equivariant d_1 Frontier

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

Six adaptive research rounds — 90 programmes
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This report executes the highest-priority problem selected in Release 11.05: classify positive face measures whose curvature energy is compatible with FIN coarse/refinement maps. The classification fixes inherited horizontal weights but proves a growing vertical nonuniqueness. A tensor-product Hodge repair exists conditionally; the next exact decision problem is the full D_{12} -equivariant second-differential classification.

Abstract

ST1992–ST2081 classify diagonal face-curvature energies under the exact FIN $12 \rightarrow 24 \rightarrow 48$ product refinement, test stronger naturality and tensor axioms, and prove that Schur/minimal-extension consistency cannot determine the new vertical face weights.

Fix a base face complex with curvature vector F and energy

$$E_0(F) = \frac{1}{2} \sum_f \kappa_f F_f^2.$$

Under $12 \rightarrow 24$, a coarse curvature embeds as two identical horizontal copies and zero curvature on every edge-times-interval square. Equality of fine and coarse energy holds if and only if

$$\kappa_{f,0}^{(24)} + \kappa_{f,1}^{(24)} = \kappa_f^{(12)}.$$

Fiber-swap symmetry uniquely sets each inherited copy to $\kappa_f/2$. All 66 new square weights remain arbitrary positive numbers; after D_{12} symmetry they form an $(\mathbb{R}_+)^6$ cone. In unreduced coordinates the 220 energy constraints leave an affine nullity of 286 among 506 fine face weights.

The classification iterates. At $24 \rightarrow 48$, inherited base faces carry $\kappa_f/4$, old square weights split into halves, and 144 new squares appear. At least seven new natural square orbits are introduced, so after two steps the vertical positive cone has dimension at least 13. Exact mod-2 boundary ranks show $H^1 = 0$ at all three product-cell levels, but no finite sequence of coarse energy equalities selects future vertical weights.

Stronger elementary axioms do not resolve the ambiguity. Positivity and joint degree-one homogeneity admit geometric, harmonic, minimum and maximum rules. Separate bilinearity in horizontal edge weight w and fiber rate q forces only the shape

$$\mu = cwq,$$

leaving c , the base face set, base measure and gauge coupling free.

The Schur test gives a sharper no-go. For one product square, write the coarse horizontal mode as a , relative horizontal mode as r , and vertical-potential difference as s . The symmetric quadratic energy is

$$E = \frac{k}{2}(a^2 + r^2) + \frac{\mu}{2}(2r + s)^2.$$

Minimizing over (r, s) gives $r = s = 0$ and

$$E_{\min}(a) = \frac{k}{2}a^2,$$

exactly independent of μ . Network-wide, identical layer connections and flat fiber links give the same result. Therefore no coarse classical energy, energy-preserving embedding or minimal-extension Schur complement can identify vertical Hodge weights. Fine holonomy data or a quantum/fluctuation measure is necessary.

A coherent conditional repair is available: given complete factor Hodge inner products and a cellular product, the tensor-product inner product gives cross-square weight proportional to wq and is associative and positive. It still inherits the unsolved base complex/measure, factor normalization, q , differential-calculus quotient and overall action scale. It is not a bootstrap from A alone.

The central no-go is therefore precise: energy preservation, positivity, automorphism naturality, homogeneity, associativity and Schur consistency do not select a unique Hodge measure. The highest-value next theorem is to classify all D_{12} -equivariant second differentials $d_1 : C^1 \rightarrow C^2$ satisfying $d_1 d_0 = 0$ and the $12 \rightarrow 24 \rightarrow 48$ refinement intertwining relations. A one-dimensional solution space after normalization would be a major breakthrough; higher dimension would be a decisive strict gauge-geometry no-go.

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1 Round I: ST1992–ST2006 — $12 \rightarrow 24$ classification

Let the selected base 2-complex have $N_2 = 220$ faces (the formulas apply to any selected face family). Its diagonal Hodge energy is

$$E_0(F) = \frac{1}{2} F^T M_0 F, \quad M_0 = \text{diag}(\kappa_f) > 0.$$

The exact binary product refinement embeds coarse curvature as

$$\iota F = (F, F, 0_{66}),$$

where the last block consists of new product squares.

Energy compatibility

$$E_1(\iota F) = E_0(F) \quad \text{for all } F$$

is equivalent coefficient by coefficient to

$$\kappa_{f,0} + \kappa_{f,1} = \kappa_f.$$

Without fiber swap, each face has a free split $(a_f \kappa_f, (1 - a_f) \kappa_f)$. Fiber-swap symmetry forces

$$\kappa_{f,0} = \kappa_{f,1} = \frac{1}{2} \kappa_f.$$

Every new square has zero curvature on the coarse image, so its positive weight is absent from the equality. Thus the complete unreduced solution space has 506 variables, 220 independent equations and nullity 286. After D_{12} and fiber-swap reduction, the inherited triangle weights are fixed but the six square-distance orbits remain:

$$(\mu_1, \dots, \mu_6) \in (\mathbb{R}_+)^6.$$

A random 220-face curvature/weight replay verifies the energy identity exactly to floating precision for arbitrary positive square weights.

Theorem 1 ($12 \rightarrow 24$ energy-compatible measure classification). *Under diagonal face energy and fiber-swap symmetry, inherited horizontal face weights are uniquely halved. The six natural vertical square-orbit weights are arbitrary positive parameters.*

2 Round II: ST2007–ST2021 — iteration and associativity

Exact product-cell enumeration gives:

Level	vertices	edges	2-cells	cycle rank
0	12	66	220	55
1	24	144	506	121
2	48	312	1156	265

Exact mod-2 boundary ranks are $(55, 121, 265)$, so the chosen product complexes have $H^1 = 0$ at all levels.

At level two, the four copies of a base face carry $\kappa_f/4$. Each old level-one square splits into two copies of weight $\mu/2$. Refining the 144 level-one edges creates 144 new squares. Their natural edge types include at least six base-distance horizontal types and one old-fiber vertical type. Consequently the symmetric vertical measure cone has dimension at least

$$6 + 7 = 13$$

after two steps.

Direct four-copy refinement and iterated binary refinement agree on horizontal quarters, so the positive inherited sector is associative. Axis exchange can relate some vertical weights only if the axes and rates are physically equivalent; the free q_1, q_2 need not be.

By induction, every nontrivial refinement introduces new edge-times-interval faces whose curvatures vanish on the prior coarse image. No finite set of coarse energy equalities can determine all future vertical weights.

3 Round III: ST2022–ST2036 — stronger axioms

For a square generated by a horizontal edge weight w and fiber rate q , all of the following rules are positive and natural:

$$wq, \quad \sqrt{wq}, \quad \frac{2wq}{w+q}, \quad \min(w, q), \quad \max(w, q).$$

The last four are jointly degree-one homogeneous and nonproportional on the six strict distance classes. Positivity, symmetry and degree-one homogeneity therefore do not select a rule.

If one adds the stronger functional axiom that the rule is separately linear in w and q , then

$$\mu(w, q) = c w q$$

is forced. This rule is associative under multiple product axes, with cross-axis faces cq_1q_2 . However, the positive constant c remains, as do the base face set, base face measure and overall Wilson coupling. The physical dimensions of w, q, μ are not supplied, so the choice of homogeneity degree is itself an assumption.

First-order factorization also has a normalization gauge: rescaling a factor 1-form inner product while compensating its differential preserves the same 0-form Laplacian but changes product 2-form normalization.

Thus bilinearity/associativity selects a useful conditional product *shape*, not a strict measure.

4 Round IV: ST2037–ST2051 — Schur no-go

For one product square set

$$a_0 = a + r, \quad a_1 = a - r,$$

and let s be the difference of the two vertical potentials. With equal horizontal copy weights, the local energy is

$$E(a, r, s) = \frac{k}{2}(a^2 + r^2) + \frac{\mu}{2}(2r + s)^2.$$

The exact stationarity equations yield

$$r = 0, \quad s = 0, \quad E_{\min}(a) = \frac{k}{2}a^2.$$

The result contains no μ .

The network generalization is immediate: copy the coarse connection to both layers and set the fiber potential flat. Every product-square curvature vanishes. The common vertical shift is a gauge zero mode and does not change the conclusion.

Theorem 2 (vertical Schur invisibility). *Under symmetric horizontal lifting, every positive vertical Hodge block gives the same coarse minimal-extension Dirichlet/curvature energy. Vertical face weights cannot be inferred from coarse classical energy or its Schur complement.*

Breaking fiber-swap symmetry can make μ enter the reduced form, but violates the earlier natural energy classification. Gaussian fluctuation determinants do depend on μ ; using them requires a path-integral measure, $\hbar$ or temperature, regularization and counterterms. Fine square holonomy or relative mode observations can also measure μ , but require an *OA* instrument.

5 Round V: ST2052–ST2066 — monoidal Hodge repair

Given complete factor 1-form Hilbert spaces and a cellular product, the tensor product inner product supplies a coherent positive cross-square measure. In a normalized convention it yields

$$\mu_e = w_e q.$$

The construction is associative and compatible with horizontal energy halving. It is the best positive conditional repair found in this cycle.

It is not a bootstrap from A alone:

- it inherits the unsolved base face set and measure;
- factor inner-product normalizations and overall coupling remain;
- q and dimensional scale remain unsourced;
- universal, clique, path and cubical calculi share d_0 but have different 2-form spaces;
- a noncommutative junk-form quotient depends on algebra representation and Dirac choice;
- spectral-triple products require complete factor triples, gradings and real structures.

Thus the monoidal Hodge family is mathematically coherent only after the very inputs the strict theory still needs to derive.

6 Round VI: ST2067–ST2081 — final classification

Object	Status	Boundary
horizontal weight halving	[Proven unique]	diagonal energy/fiber swap
vertical square weights	[Unconstrained cone]	coarse energy invisible
two-level associativity	[Proven horizontal]	vertical dimension ≥ 13
bilinear product rule cwq	[Conditional unique shape]	c /base/scale free
Schur/minimal extension selector	[Refuted]	exact μ independence
monoidal tensor Hodge	[Constructed conditional]	factor data supplied
unique strict Hodge measure	[Not established]	updated gate 0/7
continuum gauge physics	[Not established]	locality/scale/OA absent

The central theorem of the cycle is:

energy preservation fixes inherited horizontal weights but cannot fix new vertical Hodge weights.

The freedom is not a finite-level accident; it grows under iteration and lies in the kernel of the coarse curvature embedding.

7 Highest-value next theorem

Face-set and face-weight searches have reached a no-go under scalar energy compatibility. The next object must include the *second differential* itself. The decisive finite problem is:

Classify all D_{12} -equivariant maps $d_1 : C^1 \rightarrow C^2$ satisfying $d_1 d_0 = 0$, positivity of a face inner product, and the $12 \rightarrow 24 \rightarrow 48$ refinement intertwining/coherence equations. Determine the solution dimension after quotienting overall scale and basis conventions.

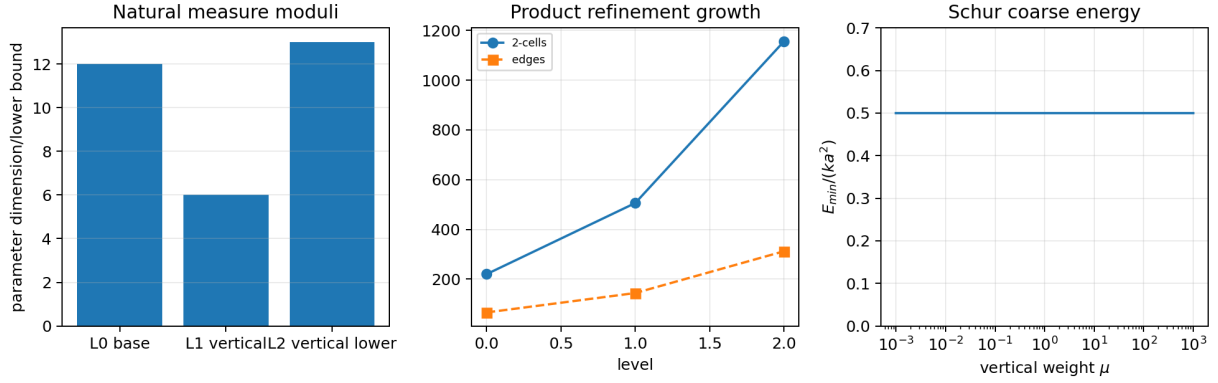


Figure 1: Left: natural measure moduli. Centre: growth of edges and 2-cells through product refinement. Right: minimized coarse square energy is constant as the vertical Hodge weight varies.

The updated gate requires: an equivariant d_1 , positive Hodge star, $12 \rightarrow 24$ chain-map identity, $24 \rightarrow 48$ associativity, uniqueness modulo one scale, continuum Maxwell scaling, and an OA record.

This problem has high decision value because it is finite linear representation theory before positivity/scaling. A one-dimensional normalized solution would be the first serious strict gauge-geometry candidate. A higher-dimensional solution space would be a decisive no-go for unique gauge geometry from current FIN data.

8 Route ranking

Route	Rigorous-success estimate	Physics decision power
D_{12} -equivariant d_1 /refinement classification	0.75	1.00
conditional monoidal product Hodge model	0.85	0.50
quantum determinant selection of vertical weights	0.30	0.65
support-only face rules	0.05	0.30
OA vertical-holonomy experiment	0.40	0.55

These are strategic judgments, not empirical probabilities.

9 Final interpretation

FIN refinement has a coherent relative cellular and monoidal Hodge extension, but coarse energy, symmetry, positivity and Schur consistency do not select its vertical gauge geometry. FIN remains a finite Dirichlet–Markov–spectral information geometry with conditional gauge extensions, not a fundamental physical theory. The equivariant second-differential classification is now the sharpest finite theorem capable of changing that verdict.

A Six-round ledger

Range	Target	Verdict
ST1992–ST2006	$12 \rightarrow 24$ face energy	horizontal halves; vertical cone

Range	Target	Verdict
ST2007–ST2021	24 $\rightarrow$ 48 associativity	quarters plus growing moduli
ST2022–ST2036	stronger axioms	product shape conditional
ST2037–ST2051	Schur/minimal extension	vertical-weight no-go
ST2052–ST2066	monoidal Hodge	coherent conditional family
ST2067–ST2081	synthesis	equivariant d_1 frontier

B Reproducibility

The authoritative result sets run from `FIN_ST1992_ST2006_Results.json` through `FIN_ST2067_ST2081_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st1992_st2081.py` verifies all ninety packets, energy identities, product-cell ranks, vertical cone counts, Schur independence, updated gate, figure and nonpromotion boundaries.

C Nonclaims

This report does not derive d_1 , a unique face measure, gauge coupling, continuum dimension, Maxwell normalization, physical scale, state ontology, platform or evidence. It does not complete legacy role transfer, QW-2191, Standard Model, gravity, L_{total} or Theory of Everything. All future FIN PDFs remain English.

References

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